

AIRPORT FLIGHT PERFORMANCE ANALYTICS

1. PROJECT PLANNING AND REPOSITORY SETUP:

16-07-2026

On the first day, the project blueprint was studied to understand the project objectives, workflow, and expected deliverables. The dataset source, SkyPath Operational Intelligence: US Flight Delay & Performance Engine, was reviewed, and its collection method, credibility, licensing, and freshness were assessed for educational use. A GitHub repository was created with organized folders (data/, sql/, notebooks/, dashboard/, and docs/) to manage the project files efficiently. Finally, the project environment was prepared, and a Day 1 work log was documented, providing a strong foundation for the remaining stages of the Airport Operations Performance Analysis project.

2. DATASET STRUCTURE AND BUSINESS MAPPING:

17-07-2026

We reviewed the business problem and defined the project success metrics. We examined the original dataset `Airport_Operations_20k_Realistic.csv` and confirmed it as the primary dataset for the project. We mapped all 35 approved dataset columns to their business meanings and prepared Data Dictionary Version 1, including column names, descriptions, and data types. This documentation provided a clear understanding of the dataset and established a strong foundation for the data profiling and cleaning stages.

3. DATASET PROFILING AND DATA QUALITY ASSESSMENT:

18-07-2026

We performed data profiling on the original dataset `Airport_Operations_20k_Realistic.csv` to understand its quality before starting the cleaning process. We first checked the total number of rows and columns to verify the dataset size. Next, we inspected the data types of all detected columns to ensure they matched the expected formats. We then analyzed each column to identify and record missing values. Finally, we checked the `Flight_ID` column for duplicate records to ensure that every flight had a unique identifier. The findings from this profiling process were documented and used as the basis for planning the data cleaning activities in the next stage.

4. DATA QUALITY VALIDATION AND PROFILING REPORT:

19-07-2026

We continued the data profiling process by identifying data quality issues in the original dataset `Airport_Operations_20k_Realistic.csv`. We examined the dataset for data type issues, such as numeric values stored as text and incorrect date formats in the

Flight_Date and Month columns. We also validated important numerical columns, including Baggage_Delay_Min, Total_Delay_Min, and Passenger_Count, to identify invalid or out-of-range values. Next, we reviewed categorical columns like Aircraft_Type and Status to detect inconsistent category names, spelling variations, and formatting differences. Finally, all identified data quality issues were documented in a Data Profiling Report, which served as the reference for preparing the data cleaning plan in the next phase.

5. DATA CLEANING STRATEGY AND PLANNING:

20-07-2026

We prepared a comprehensive data cleaning plan based on the issues identified during the data profiling stage. We defined appropriate cleaning methods for handling missing values, removing duplicate records, correcting incorrect data types, and standardizing inconsistent category values. For each affected column, we decided whether the data should be kept, removed (drop), or imputed based on its quality and business relevance. After finalizing all cleaning rules, we documented the complete Cleaning Plan Version 1 and saved it in the /docs folder. This document serves as the official guideline for implementing the data cleaning process consistently across Excel, SQL, and Python.

6. DATA CLEANING IN MICROSOFT EXCEL:

21-07-2026

We implemented the approved data cleaning plan using Microsoft Excel on the original dataset Airport_Operations_20k_Realistic.csv. We opened the dataset and applied filters and conditional formatting to identify missing values, duplicates, and inconsistent records. We corrected incorrect data types, removed extra spaces using the TRIM function, and standardized the Aircraft_Type and Status columns to ensure consistent category names. After completing the cleaning process, we validated the cleaned dataset by comparing the row count with the original dataset to ensure that only intended changes were made. The cleaned dataset was then saved for use in the next stages of SQL, Python, and Power BI analysis.

7. EXCEL DATA ANALYSIS AND BUSINESS INSIGHTS:

22-07-2026

We analyzed the cleaned dataset using Microsoft Excel to understand airport operational performance. We created Pivot Tables by grouping the data based on Aircraft_Type and Status, and calculated key metrics such as Baggage_Delay_Min, Total_Delay_Min, and Passenger_Count. Using these summaries, we prepared initial Excel reports and identified 3 to 5 business insights related to flight delays, passenger traffic, and aircraft performance. After completing the analysis, we saved and exported the cleaned Excel workbook to the project repository for future reference and further analysis using SQL, Python, and Power BI.

8. SQL DATABASE SETUP AND DATA CLEANING:

23-07-2026

We imported the original dataset `Airport_Operations_20k_Realistic.csv` into the SQL database and created the required table structure. We defined `Flight_ID` as the primary key to uniquely identify each flight record. Next, we reproduced the data cleaning process in SQL by handling missing values, removing duplicate records, and correcting data types using SQL queries. After completing the cleaning process, we created and saved the database structure in `schema.sql`. This ensured that the dataset was clean, well-structured, and ready for SQL analysis and KPI generation.

9. SQL ANALYSIS AND KPI DEVELOPMENT:

24-07-2026

We developed SQL queries to analyze the cleaned airport operations data. We created the required lookup joins to combine related information and improve data analysis. Next, we performed aggregation queries by grouping the data based on `Aircraft_Type` and `Status` to calculate important metrics such as `Baggage_Delay_Min`, `Total_Delay_Min`, and `Passenger_Count`. Finally, we wrote KPI queries based on the project blueprint to measure key business metrics, including total flights, average delays, passenger count, and airport operational performance. These SQL queries formed the foundation for reporting and Power BI dashboard development.

10.ADVANCED SQL ANALYSIS AND DOCUMENTATION:

25-07-2026

On Day 10, we created SQL Views for the most important Key Performance Indicators (KPIs) to simplify reporting and improve query reusability. We applied SQL Window Functions such as `Running Total`, `RANK()`, `LAG()`, and `LEAD()` on the `Flight_Date` and `Month` columns to analyze trends and compare flight performance over time. Finally, we organized all SQL scripts, including the schema, cleaning queries, KPI queries, and views, into the `sql/` folder and prepared a `README` file explaining the purpose and usage of each SQL file. This completed the SQL implementation phase and prepared the project for Python analysis and Power BI dashboard development.

11.DATA CLEANING VALIDATION USING PYTHON (PANDAS):

26-07-2026

We reproduced the complete data cleaning process using Python and the Pandas library. We loaded the original dataset into a Jupyter Notebook and applied the same cleaning steps that were previously performed in Excel and SQL, including handling missing values, removing duplicate records, correcting data types, and standardizing categorical values. After cleaning, we validated the dataset by comparing the row counts and null value counts with the approved cleaning plan to ensure consistency across all platforms. Finally, we documented the entire cleaning process and committed the notebook as `01_clean.ipynb` to the `notebooks/` folder in the project repository.

12.EXPLORATORY DATA ANALYSIS (EDA) USING PYTHON:

27-07-2026

We performed Exploratory Data Analysis (EDA) on the cleaned airport operations dataset using Python. We analyzed the distribution and summary statistics of `Baggage_Delay_Min`, `Total_Delay_Min`, and `Passenger_Count` to understand their overall patterns and variability. We also conducted segment analysis by grouping the data based on `Aircraft_Type` and `Status` to compare operational performance across different categories. In addition, we extracted useful date components such as `Year`, `Month`, and `Day` from the `Flight_Date` column to support time-based analysis. The results of this analysis provided a better understanding of the dataset and prepared it for advanced statistical analysis and visualization.

13.OUTLIER DETECTION AND DATA VISUALIZATION:

28-07-2026

We performed outlier analysis on the `Baggage_Delay_Min`, `Total_Delay_Min`, and `Passenger_Count` columns using the Interquartile Range (IQR) and Z-score methods to identify unusual values that could affect the analysis. We then created four different charts to visualize the dataset, including distribution, trend, comparison, and composition charts, which helped us better understand airport operations and flight performance. Finally, we analyzed the results and documented five key dataset-based insights, highlighting important patterns related to delays, passenger traffic, aircraft types, and flight status. These findings provided meaningful business insights and supported the next stage of statistical analysis and dashboard development.

14.STATISTICAL ANALYSIS AND DISTRIBUTION ASSESSMENT:

29-07-2026

We performed descriptive statistical analysis on the `Baggage_Delay_Min`, `Total_Delay_Min`, and `Passenger_Count` columns to calculate key statistical measures such as count, mean, median, minimum, maximum, standard deviation, and quartiles. We then analyzed the data distributions by creating histograms and calculating skewness and kurtosis to understand the shape, spread, and variability of each variable. Finally, we documented the sampling caveats, including assumptions, dataset limitations, and potential biases that could influence the analysis results. This statistical assessment helped validate the quality of the dataset and provided a strong foundation for the final business insights and dashboard development.

15.CORRELATION ANALYSIS AND STATISTICAL VALIDATION:

30-07-2026

We performed a correlation analysis to measure the relationship between `Baggage_Delay_Min`, `Total_Delay_Min`, and `Passenger_Count` using a correlation matrix. We then conducted variance analysis by comparing different `Aircraft_Type` and `Status` categories to understand variations in airport operational performance. Finally,

we validated the project's main hypothesis using statistical analysis, interpreted the results, and documented the key findings. We also recorded the limitations of the analysis, including assumptions, data constraints, and factors that may influence the interpretation of the results. This completed the statistical analysis phase and prepared the dataset for Power BI dashboard development and business reporting.

16. POWER BI DATA MODELING AND KPI DEVELOPMENT:

31-07-2026

We imported the cleaned airport operations dataset into Power BI Desktop and verified that the data was loaded successfully. We created the required data model and established relationships between the tables using appropriate key columns to ensure accurate analysis. Next, we developed DAX measures based on the project blueprint to calculate important business KPIs, such as Total Flights, Total Passengers, Average Baggage Delay, Average Total Delay, Flight Delay Percentage, and On-Time Performance. Finally, we designed KPI cards to display the top 4–6 business metrics on the dashboard, providing a clear and interactive overview of airport operational performance.

17. POWER BI DASHBOARD DESIGN:

01-08-2026

We designed the main Power BI dashboard pages to present the airport operations analysis in a clear and interactive format. We created separate pages for Overview, Trends, and Segments to display overall performance, time-based analysis using Flight_Date and Month, and category-wise analysis using Aircraft_Type and Status. We added filters and slicers to allow users to interact with the dashboard and analyze specific aircraft types and flight statuses. Finally, we implemented drill-down functionality, enabling users to navigate from summary-level KPIs to detailed records for deeper analysis. This improved the dashboard's usability and supported better business decision-making.

18. ENHANCEMENT AND PROJECT FINALIZATION:

02-08-2026

We enhanced the Power BI dashboard by applying business storytelling techniques to make the insights more meaningful and easy to understand. We added clear titles, annotations, and callouts to highlight important KPIs and business findings. We then improved the overall dashboard design by refining the theme, adjusting spacing and alignment, customizing tooltips, and performing an accessibility check to ensure a user-friendly experience. Finally, we exported the completed .pbix file and captured dashboard screenshots, which were saved in the project repository for documentation, presentation, and future reference.

19.PROJECT CONCLUSION AND FINAL DOCUMENTATION:

03-08-2026

We prepared the final business insights and recommendations based on the KPIs developed throughout the project. We interpreted the dashboard results and provided recommendations to improve airport operational efficiency, reduce flight delays, and enhance passenger experience. We also documented the limitations of the dataset, including assumptions, data quality constraints, and factors that could affect the analysis. Finally, we prepared a data cleaning log that recorded the complete cleaning process, including before and after row counts, duplicate removal, missing value handling, data type corrections, and category standardization. This final documentation summarized the entire project and completed the Airport Operations Performance Analysis.

20.PROJECT DOCUMENTATION AND PORTFOLIO PREPARATION:

04-08-2026

We completed the final project documentation by reviewing and updating the Data Dictionary with accurate column descriptions and data types. We finalized the README file by including the project overview, objectives, tools used, dataset information, project workflow, and execution steps. We organized and committed all SQL scripts and Python notebooks, adding clear comments to explain the purpose of each file. Next, we prepared the Power BI dashboard documentation, describing the DAX measures, KPI definitions, and the purpose of each dashboard page. Finally, we assembled all project artefacts, including the cleaned dataset, SQL files, Python notebooks, Power BI dashboard, documentation, screenshots, and GitHub repository, creating a complete portfolio ready for project submission and professional presentation.

21.PROJECT PRESENTATION AND FINAL SUBMISSION:

05-08-2026

We prepared and rehearsed the final project presentation, covering both the business objectives and the technical implementation of the Airport Operations Performance Analysis project. We organized the presentation to explain the business problem, dataset, data cleaning process, SQL analysis, Python-based statistical analysis, and Power BI dashboard in a logical sequence. We also prepared for the technical defence by reviewing the data cleaning methodology, SQL queries, statistical analysis, KPI calculations, DAX measures, and dashboard design to confidently answer project-related questions. Finally, we completed the project submission by providing links to the GitHub repository, Power BI dashboard (.pbix), and project portfolio, ensuring that all documentation, source files, and project artefacts were properly organized and ready for evaluation.